INCH-POUND

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DETAIL SPECIFICATION

STEEL FORGINGS

REACTIVATED AFTER 21 JULY 2025 AND MAY BE USED FOR NEW AND EXISTING DESIGN AND ACQUISITIONS.

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers steel forgings that are produced by a hot deformation process such as hammering, pressing, upsetting, extruding, or any combination of these processes (see 6.1).

1.2 Classification. The forgings are classified for inspection purposes and should be ordered (see 6.2) on the basis of grade and condition.

1.2.1 Grade.

1.2.1.1 Grade A forgings. For grade A forgings, mechanical properties are to be measured in the body of the forging (test coupon - see 6.7.7) as well as in the integrally forged test metal (see 6.7.8) or the prolongation (see 6.7.4) at the locations shown on the applicable drawing. Correlation should be determined by use of a value difference (see 6.7.9) between the properties of the test coupon of the forging and those of its integrally forged test metal or the prolongation.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory (ARL), Army Research Directorate, Specifications & Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A AREA FORG

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1.2.1.2 Grade A1 forgings. For grade A1 forgings, mechanical properties should be measured in the body of the forging (test coupon - see 6.7.7) as well as in the integrally forged test metal (see 6.7.8) or the prolongation (see 6.7.4) at the locations shown on the applicable drawing. Correlation should be determined by use of a value difference (see 6.7.9) between the properties of the test coupon and those of its integrally forged test metal or the prolongation. When the test metal or prolongation of a forging is not tested for mechanical properties, that forging may be accepted on the basis of hardness.

1.2.1.3 Grade B forgings. For grade B forgings, mechanical properties should be measured in the integrally forged test metal or the prolongation from the forging at the locations shown on the applicable drawing.

1.2.1.4 Grade C forgings. For grade C forgings, mechanical properties of the forging should be determined directly from the body of the forging (test coupon - see 6.7.7) from a selected or representative forging. Forgings not sectioned for testing may be measured on the basis of hardness.

1.2.1.5 Grade D forgings. For grade D forgings, mechanical properties of the forging are measured on the basis of hardness only.

1.2.2 Condition. Forgings should be furnished in the normalized, annealed, or normalized and tempered condition unless mechanical properties are specified on the applicable drawing or in the contract or purchase order (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product
MIL-STD-2073-1	-	Standard Practice for Military Packaging

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

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THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at https://ansi.org/).

ASTM INTERNATIONAL

ASTM A275/A275M	-	Magnetic Particle Examination of Steel Forgings
ASTM A370	-	Standard Test Methods and Definitions for Mechanical Testing of Steel Products
ASTM A751	-	Methods, Practices, and Definitions for Chemical Analysis of Steel Products
ASTM D3951	-	Practice for Commercial Packaging
ASTM E8/E8M	-	Standard Test Methods for Tension Testing of Metallic Materials
ASTM E10	-	Brinell Hardness of Metallic Materials
ASTM E18	-	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
ASTM E23	-	Notched Bar Impact Testing of Metallic Materials
ASTM E340	-	Method for Macroetching Metals and Alloys
ASTM E381	-	Standard Method of Macroetch Testing Steel Bars, Billets, Blooms, and Forgings
ASTM E390	-	Standard Reference Radiographs for Steel Fusion Welds

(Copies of these documents are available online at https://www.astm.org).

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1	-	Structured Welding Code

(Copies of these documents are available online at https://pubs.aws.org/).

SAE INTERNATIONAL

SAE AIR4127	-	Steel: Chemical Composition and Hardenability
AMS 2301	-	Aircraft Quality Steel Cleanliness - Magnetic Particle Inspection Procedure

(Copies of these documents are available online at https://www.sae.org/.)

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2.4 Order of precedence. Unless otherwise noted herein or within the contract or purchase order, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. Steel forgings shall be in accordance with the applicable drawings, the requirements of this specification, and all referenced documents. Any conflicts are to be resolved by means of the order of precedence (see 2.4).

3.2 First article. The contractor shall submit a first article sample unless it is specifically waived in the contract (see 6.2), by the responsible engineering activity and approval by the responsible product assurance activity (see 6.4).

3.2.1 First article residual metal. Residual metal from the first article units shall be retained until the acceptance of the first article (see 6.2).

3.3 Chemical composition.

3.3.1 Chemical composition specified on drawing. When chemical composition is specified on the drawing in the invitation for bids, contract, or purchase order, the composition shall be verified on the basis of ladle analysis, the chemical limits where the declared chemistry shall be submitted to the contracting agency or its authorized representative. The chemical composition shall be determined by a product analysis (values shall be listed as weight percent) in accordance with 4.6.4.6 and 4.6.7.1. If an analysis of Antimony, Arsenic or Lead is required, it shall be specified in the contract or purchase order (see 6.2).

3.3.2 Chemical composition not specified on drawing. The contractor shall select a steel composition which when heat treated shall meet the mechanical properties specified on the applicable drawing and when specified in the contract. The chemistry of the steel composition shall be verified on the basis of ladle analysis and the chemical limits thereof shall be subject to check analysis tolerances in accordance with SAE AIR4127.

3.3.3 Certification of chemical analysis. The contractor shall furnish a certified chemical analysis of each melt of steel used in producing the forgings showing the percentage of carbon, sulfur, phosphorus, silicon and manganese (see 6.4). For alloy steels, in addition to these elements, the percentage of the alloying elements shall be shown.

3.4 Material. Forgings shall be produced from cast ingots and fabricated stock and conform to the chemical and mechanical property requirements of the detail specifications, drawings, and the contract or purchase order. Materials shall be worked prior to and during the forging operation to such extent as to produce a wrought metallurgical structure.

3.5 Mechanical properties.

3.5.1 Required property attributes. The required mechanical property attributes for the different grades of forgings shall be as shown in table I.

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TABLE I. Forging grades and mechanical property attributes.

Mechanical Property Attributes	Requirement Type	Grade				
		A	A1	B	C	D
Test Coupon	FA [1/]	Yes	Yes	No	Yes	No
	P [2/]	Yes	Yes	No	Yes	No
Test Metal	FA [1/]	Yes	Yes	Yes	No	No
	P [2/]	Yes	Yes	Yes	No	No
Value Difference	FA [1/]	Yes	Yes	No	No	No
	P [2/]	Yes	Yes	No	No	No
Yield Strength	FA [1/]	Yes	Yes	Yes	Yes	No
	P [2/]	Yes	Yes	Yes	Yes	No
Reduction of Area	FA [1/]	Yes	Yes	Yes	Yes	No
	P [2/]	Yes	Yes	Yes	Yes	No
Charpy Impact Resistance	FA [1/]	Yes	Yes	Yes	Yes	No
	P [2/]	Yes	Yes	Yes	Yes	No
Hardness	FA [1/]	Yes	Yes	Yes	Yes	Yes [4/]
	P [2/]	Yes [3/]	Yes [3/]	Yes [4/]	Yes [3/]	Yes [4/]

[1/] First article.
[2/] Production.
[3/] Production forgings may be evaluated by values obtained from same surface-area-of body of forging where test coupons are taken for evaluating each forging lot.
[4/] Hardness.

3.5.2 Mechanical property test specimens. Test specimens for mechanical properties shall be taken from the applicable test coupon(s) and test metal at the specified locations thereof.

3.5.3 First article.

3.5.3.1 Test Coupon. When a test coupon (see 6.7.7) is required, the value obtained for each of the yield strength, reduction of area, and Charpy impact resistance attributes for each respective test specimen shall be in compliance with the requirements of table II for the yield strength specified on the applicable engineering drawing.

3.5.3.2 Test metal. When test metal (see 6.7.8) is required, the value obtained for each of the yield strength, reduction of area and Charpy impact resistance attributes for each respective test specimen shall be in compliance with the requirements of table II for the yield strength specified on the applicable engineering drawing.

3.5.3.3 Value difference. When required, a value difference (see 6.7.9) shall be calculated for each of the following three mechanical property attributes of the forging: yield strength, reduction of area, and Charpy V-notch impact resistance.

3.5.4 Production article. For each of the forgings inspected which requires an adjusted value (see 6.7.10), the adjusted value for each of the yield strength, reduction of area, and Charpy impact resistance attributes of the test metal shall be in compliance with the requirements of table II for the yield strength specified on the applicable engineering drawing. For those forgings that do not require an adjusted value, all the applicable requirements listed in this specification for that forging grade shall be complied with (see table I).

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TABLE II. Mechanical properties.

Yield strength (0.10 percent offset)	Minimum reduction of area (single value) Percent (%)		Minimum Charpy V-notch impact resistance at -40F ± 2F (Single value foot-pounds)	
	Transverse	Longitudinal	Transverse	Longitudinal
60,000 - 69,999	40	50	55	85
70,000 - 79,999	40	50	50	80
80,000 - 89,999	40	50	45	70
90,000 - 99,999	40	50	40	60
100,000 - 109,999	40	50	35	50
110,000 - 119,999	40	50	35	50
120,000 - 129,999	40	50	35	45
130,000 - 139,999	35	45	30	40
140,000 - 149,999	30	40	25	35
150,000 - 159,999	30	40	20	30
160,000 - 169,999	25	35	15	25
170,000 - 180,000	25	35	15	20

3.5.5 Mechanical property requirements. When required, the mechanical properties of the forgings shall conform to the following requirements.

3.5.5.1 Yield strength. When required, yield strength shall be as specified on the applicable drawings, in the contract, or in the order.

3.5.5.2 Reduction of area. When required, the minimum percent reduction of area for individual specimens shall be as specified in table II or as specified on the applicable drawing.

3.5.5.3 Charpy V-notch impact resistance. When required, the minimum value for individual Charpy V-notch impact resistance specimens shall be as specified in table II for the average yield strength or as specified on the applicable drawing.

3.5.5.4 Hardness. When required, hardness values for any forgings shall be as specified on the applicable engineering drawing or in the contract or order.

3.5.5.5 Transverse properties. Unless otherwise specified on an applicable engineering drawing, mechanical tests shall be taken in the transverse direction, i.e., such that the lengthwise axis of the test specimen is at right angles to the grain flow of the forging. When the thickness of the sample or the forging prevents the preparation of transverse specimens, longitudinal specimens for the determination of yield strength, reduction of area, or impact resistance shall apply.

3.6 Process controls.

3.6.1 Heat treatment. The forgings shall be given a uniform heat treatment at temperatures necessary to produce material that will conform to the requirements of this specification. When mechanical properties are specified (see 1.2.2 and 6.2), the minimum heat treat process shall consist of normalizing, austenitizing, quenching in an agitated liquid,

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and tempering. The heat treatment shall be applied to the whole of the forging and never to a portion only, except when localized heat treating is specified in the drawing.

3.6.2 Repair welding. Unless approved by the contracting officer prior to repairing forgings, forgings shall not be repaired by welding. When welding is permitted by the contracting officer, with concurrence of the applicable cognizant engineering authority and responsible product assurance directorate, it shall be done prior to final heat treatment. The contractor's request shall include all information necessary for analysis and decision (see 6.4). The responsible development engineer shall assist the contracting officer and quality control in making the repair weld decision.

3.6.2.1 Test plate. The mechanical properties of the welded test plate (see 6.4) shall be submitted with the request for repair welding. The test plate shall be prepared in accordance with paragraph 5.19 of AWS D1.1 titled "Groove Weld Plate Qualification Test for Plate of Limited Thickness". The side bend specimen call-out shall be disregarded. The welded test plate shall be the same material as the forging and shall have undergone the same heat treatment process as the forging. The test plate shall be of such size that three transverse tension specimens and three transverse Charpy V-notch impact specimens can be obtained. These specimens shall be tested as specified in ASTM E8/E8M and ASTM E23, respectively. The mechanical properties of the specimens shall be in accordance with the applicable drawing.

3.7 Material soundness.

3.7.1 Surface. All forgings shall be free from cracks, forging laps, seams, and shuts.

3.7.2 Internal. When specified in the contract or purchase order (see 6.2), the soundness of steel materials used in forgings shall be determined by macroetching. For acceptance, the material shall comply with the requirements of ASTM E381 as follows:

a. Carbon steel - Equal to or better than S-3, R-2, C-3 inclusive shall be acceptable.

b. Alloy steel

36 square inches or less:

Acceptable to S-3 inclusive
to R-2 inclusive
to C-2 inclusive

Over 36 square inches:

Acceptable to S-3 inclusive
to R-3 inclusive
to C-3 inclusive

3.7.3 Cleanliness. When cleanliness is specified (see 6.2), bars and billets shall conform to the requirements of AMS 2301.

3.7.4 Wake bar. No wake bars shall be allowed for retesting for grades A, Al, B, and C forgings except those allowed by 4.7.1.1 and 4.7.1.2. Wake bars are not necessary for grade D forgings.

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3.8 Metallographic requirements. When specified in the invitation for bids, contract, or order, the austenitic grain size and metal flow patterns shall conform to the requirements of the applicable engineering drawings, if stated thereon.

3.9 Magnetic particle inspection. Unless otherwise specified in the invitation for bids, contract, or order, the forgings shall be subject to magnetic particle inspection. No cracks are acceptable. Cracks include heat treating cracks, shrink, laps, flakes, and bursts.

3.10 Dimensions and tolerances. Dimensions and dimensional tolerances shall be as specified on the applicable engineering drawing.

3.10.1 Draft angle. The draft angle shall be seven degrees maximum. Draft angle is additive to feature tolerance. Any matching draft angle shall be acceptable.

3.11 Surface roughness. Surface roughness shall be as specified on the applicable drawing.

3.12 Identification marking. Forgings shall be marked as specified on the drawing and in accordance with MIL-STD-130 and shall include the manufacturer's identification number and heat number or a heat code. When a heat code is used, a cross reference to the heat number shall appear on the test reports and packing slips.

3.13 Records. The contractor shall maintain records of the chemical and physical properties for forging analysis (see 6.4) with the period of record retention specified in the contract or purchase order.

3.14 Workmanship. The forging shall not contain defects which, due to their nature, degree, or extent, prevent the fulfillment of the requirements of this specification. The existence of defects which cannot be removed by machining the forging to its finished dimensions shall cause the forging to be rejected even if all other requirements of the specification are met.

4. VERIFICATION

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection system or quality program. The inspections set forth in the specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of their responsibility for assuring that all supplies submitted to the Government for acceptance conform to all requirements of the contract or purchase order.

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4.2 Quality assurance terms and definitions. The quality assurance terms and definitions used in this specification are in accordance with MIL-STD-109.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.5).
b. Quality conformance inspection (see 4.6).

4.4 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in the applicable test method document or paragraph(s) in the specification.

4.5 First article inspection. A first article shall be submitted for inspection in accordance with contract requirements (see 6.2.d). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to all inspection specified herein (see 4.5.2 and 4.6) and such other inspections as is necessary to determine that all the requirements of the contract have been met.

4.5.1 First article. Unless otherwise specified in the contract or purchase order, (see 6.2.d), the first article shall consist of a minimum of one forging.

4.5.2 Inspection provisions.

4.5.2.1 Grade A and Al forgings. Mechanical property tests: yield strength, reduction of area, Charpy impact resistance, and hardness; shall be conducted on the test coupon (see 6.7.7) obtained by sectioning the forging. The same tests shall be conducted on the test metal (see 6.7.8) of the forging. The mechanical properties shall comply with 3.5.3.1, 3.5.3.2, and 3.5.5.4. A value difference (see 6.7.9) shall be computed for each of the following mechanical properties: yield strength, reduction of area, and Charpy impact resistance.

4.5.2.2 Grade B forgings. Mechanical property tests: yield strength, reduction of area, Charpy impact resistance, and hardness shall be conducted on the test metal (see 6.6.7) of the forging. The mechanical properties shall comply with 3.5.3.2 and 3.5.5.4.

4.5.2.3 Grade C forgings. Mechanical property tests: yield strength, reduction of area, Charpy impact resistance, and hardness shall be conducted on the test coupon (see 6.6.6), obtained by sectioning the forging. The mechanical properties shall comply with 3.5.3.1 and 3.5.5.4.

4.5.2.4 Grade D forgings. Mechanical property tests for hardness shall be conducted on the surface of the forgings. The mechanical properties shall comply with 3.5.5.4.

4.6 Quality conformance inspection. Quality conformance inspections shall be as specified in table III.

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TABLE III. Quality conformance inspection.

Characteristic	Requirement Paragraph	Verification Paragraph	Test Method Paragraph
Chemical Composition	3.3	4.6.4.6	4.6.7.1
Dimensional	3.10	4.6.7.9	4.6.7.9
Weld repair	3.6.2	4.6.5.1	4.6.7.6, 4.6.7.6.1
Surface soundness	3.7.1	4.6.5.2.1	4.6.5.2.1
Internal soundness	3.7.2	4.6.5.2.2	4.6.7.7
Metallographic	3.8	4.6.5.2.2.1	4.6.7.7
Cleanliness	3.7.3	4.6.5.3	4.6.7.8
Identification marking	3.12	4.6.5.4	4.6.5.4
Packaging	5.1, 6.6	4.6.5.6	4.6.5.6
Workmanship	3.14	4.6.5.5	4.6.5.5
Yield strength	3.5.4, 3.5.5.1	4.6.4	4.6.7.2
Reduction in area	3.5.4, 3.5.5.2	4.6.4	4.6.7.3
Charpy impact resistance	3.5.4, 3.5.5.3	4.6.4	4.6.7.4
Hardness	3.5.5.4	4.6.4	4.6.7.5

4.6.1 Inspection lot. An inspection lot shall consist of all forgings that are produced from the same individual melt of steel and produced to the same drawing; and that, if heat treated, are heat treated together by batch procedures or being passed consecutively and continuously through a continuous type of heat treatment process.

4.6.2 Mechanical property acceptance testing. In the event test metal (see 6.7.8) or coupon metal (see 6.7.7) is of insufficient size to accommodate at least two specimens each, additional forgings may be tested as directed by the procuring activity.

4.6.2.1 Sample forgings. Forgings (see 6.7.1) that require mechanical testing, but which are supplied in other than the final heat-treated condition, shall be sampled as directed by the procuring activity. The sample forgings (see 6.7.1) shall be given the final heat treatment and shall be tested to demonstrate the capability of obtaining the specified mechanical properties.

4.6.2.2 Test metal removal. Unless otherwise specified, test metal which is not used in quality conformance testing shall be removed before shipment of forgings.

4.6.2.3 Test values. When a value difference (see 6.7.9) is calculated, transverse and longitudinal test values shall not be intermixed.

4.6.2.4 Production change. A new set of value differences (see 6.7.9), shall be calculated and used whenever the alloy system or grade has been changed during production or whenever the heat treat procedure has been changed during production.

4.6.3 Sampling. Unless otherwise specified in the contract or purchase order or directly in this specification, the formation, size, and presentation of lots for acceptance inspection shall be in be established in accordance with MIL-STD-1916 and conducted in accordance with ANSI/ASQ Z1.4 and shall be the responsibility of the contractor. The Government reserves the right to disapprove in its entirety, or in part, the contractor's method of presentation.

4.6.3.1 For chemical composition. One sample shall be selected from each melt of steel represented in the lot for comparison with the ladle analysis.

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4.6.3.2 For dimensions, tolerances, and surface roughness. In an inspection lot of ten forgings or less, each forging shall be inspected. In an inspection lot of more than ten forgings, ten forgings plus one forging from each successive group of ten forgings shall be inspected. Dimensions, tolerances, and surface roughness shall be measured for conformance to the applicable drawing.

4.6.3.3 For mechanical properties. When mechanical properties are specified, the sampling plans (including 100% inspection) far grade A, grade Al, grade B, grade C, and grade D forgings shall be as follows:

4.6.3.3.1 Grade A. One hundred percent inspection (see 4.6.4.1).

4.6.3.3.2 Grade A1. The test metal (se 6.7.8) of 20 percent of the forgings in each inspection lot shall be tested for mechanical properties. For lot sizes of ten forgings or less, the test metal of two forgings shall be tested (see 4.6.4.2).

4.6.3.3.3 Grade B. One hundred percent inspection (see 4.6.4.3).

4.6.3.3.4 Grade C. The test coupons (see 6.7.7) from a minimum of two (2) forgings in each inspection lot shall be tested for mechanical properties (see 4.6.4.4)

4.6.3.3.5 Grade D. Twenty percent of the forgings in each inspection lot shall be hardness tested (see 4.6.4.5).

4.6.4 Tests.

4.6.4.1 Grade A forgings. The test metal (see 6.7.7) of each forging shall be tested for the mechanical properties: yield strength, reduction of area, Charpy impact resistance, and hardness. By application of value differences (see 6.7.9), computed in 4.5.2.1, adjusted values (see 6.7.10) shall be calculated for yield strength, reduction of area, and Charpy impact resistance, and shall comply with 3.5.4 for each forging. Hardness testing of the test metal of each forging shall be conducted in accordance with 4.6.7.5 and shall consist of two indentations. Individual hardness values of the test metal of each forging shall comply with 3.4.5.4. Failure of each forging to comply with both requirements (3.4.4 and 3.4.5.4) shall be cause for rejection of the forging.

4.6.4.2 Grade Al forgings. Sampling for mechanical properties of each inspection lot shall be as specified in 4.6.3.3.2. The test metal (see 6.7.7) of each forging in the sample shall be tested for the mechanical properties: yield strength, reduction of area, Charpy impact resistance, and hardness. By application of value differences (see 6.7.9), computed in 4.5.2.1, adjusted values (see 6.7.10) shall be calculated for yield strength, reduction of area, and Charpy impact resistance, and shall comply with test metal of each forging in the sample. Hardness testing shall be conducted in accordance with 4.6.7.5 and shall consist of two indentations. Individual hardness values of the test metal of each forging in 3.4.4 for each forging in the sample. Hardness testing of the sample shall comply with 3.5.5.4. Lot acceptance criterion shall require that each forging in the sample comply with 3.5.4 and 3.5.5.4. In addition to the preceding tests, a hardness comparison test, conducted in accordance with 4.6.7.5 and consisting of two indentations, shall be performed on the forging surface at different, but approximately the same relative locations on each forging: for instance, one on the top surface and one on the bottom surface if possible, and in the lot. Lot acceptance criterion 2 shall require that the average hardness of each forging not in the sample

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shall be within plus or minus 25 Brinell hardness numbers, or the equivalent thereof, of the average hardness of the sample. Failure to comply with both lot acceptance criteria shall be cause for rejection of the inspection lot.

4.6.4.3 Grade B forgings. The test metal (see 6.7.8) of each forging shall be tested for the mechanical properties: yield strength, reduction of area, Charpy impact resistance, and hardness. Mechanical properties shall comply with 3.5.5. Failure to comply shall be cause for rejection of the forging.

4.6.4.4 Grade C forgings. Destructive sampling for mechanical properties shall be as specified in 4.6.3.3.4. The test coupons (see 6.7.7) of the forgings in the sample (which are obtained by sectioning the forgings) shall be tested for mechanical properties: yield strength, reduction of area, Charpy impact resistance, and hardness. Hardness testing of the test coupons of each forging in the sample shall be conducted in accordance with 4.6.7.5 and shall consist of two indentations. Lot acceptance criterion 1 shall require that the mechanical properties: yield strength, reduction of area, Charpy impact resistance, and individual hardness values of each forging in the sample shall comply with 3.5.5. In addition to the preceding tests, a hardness comparison test, conducted in accordance with 4.6.7.5 and consisting of two indentations, shall be performed on the forging surface at different, but approximately the same relative locations on each forging: for instance, one on the top surface and one on the bottom surface if possible, in the lot. Lot acceptance criterion 2 shall require that the average hardness of each forging not in the sample shall be within plus or minus 25 Brinell hardness numbers, or the equivalent thereof, of the average hardness of the sample. Failure to comply with both lot acceptance criteria shall be cause for rejection of the inspection lot.

4.6.4.5 Grade D forgings. Sampling for mechanical properties shall be as specified in 4.6.3.3.5. A hardness test, conducted in accordance with 4.6.7.5 and consisting of two indentations shall be performed on the surface of each forging in the sample at approximately the same locations. Each individual hardness value in the sample shall comply with 3.5.5.4. Failure of any forging in the sample to comply shall be cause for rejection of the represented inspection lot.

4.6.4.6 Chemical composition. The chemical composition shall be determined for each heat of steel used in making the forgings for MIL-DTL-46172B compliance with the requirements of 3.3 using the method of 4.6.7.1.

4.6.5 Examinations.

4.6.5.1 Weld repair. Each forging which has been repaired by welding shall be examined for compliance with 3.5.2 by the method of 4.6.7.6.

4.6.5.2 Material soundness.

4.6.5.2.1 Surface. Unless otherwise specified, the surfaces of each forging shall be visually examined for compliance with 3.7.1. Unless otherwise specified in the contract or order, the forging shall be subjected to magnetic particle inspection in accordance with ASTM A275/A275M to determine compliance with 3.9.

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4.6.5.2.2 Internal. When specified in the contract or order, the soundness of steel material used for the forgings shall be determined by macroetching to comply with the requirements of 3.7.2.

4.6.5.2.2.1 Metallographic. When specified in the contract or order, the forging section shall be subjected to macroetching to determine compliance with 3.8.

4.6.5.3 Cleanliness. When specified in the contract or order, the cleanliness of the steel material used for the forgings shall be subjected to inspection per AMS 2301.

4.6.5.4 Identification marking. Forgings shall be examined visually to determine compliance with marking requirements of 3.12.

4.6.5.5 Workmanship. Forgings shall be visually examined to determine compliance with the requirements of 3.14. Forgings displaying evidence of poor workmanship, such as the presence of seams, laps, injurious segregation, cracks, flakes, etc., and the presence of burrs, nicks, fins, etc. shall be cause for rejection.

4.6.5.6 Packaging. Packaging shall be examined visually and in accordance with MIL-STD-2073-1 to determine compliance with the packaging requirements of Section 5.

4.6.6 Test Specimens.

4.6.6.1 Tension specimens. Unless otherwise specified, tension test specimens shall be machined to the form and dimensions of the 0.500 (0.505) inch, 0.350 (0.357) inch, or 0.250 (0.252) inch diameter specimen as shown in ASTM E8/E8M. Diameters in parenthesis refer to NOTE 6 of FIG. 8 titled “Standard 0.500-in. (12.5mm) Round Tension Test Specimen with 2-in. (50mm) Gage Length”, of ASTM E8/E8M.

4.6.6.2 Charpy impact specimens. Three (3) Charpy V-notch impact specimens each shall be machined to the form and dimensions of the standard 0.394 inch by 0.394 inch by 2.165 inch specimen in accordance with ASTM E23 and ASTM A370 for both the transverse (TL) orientation and longitudinal (LT) orientation from each sample as directed under 3.5.3.1 and defined under 6.7.8. The notch shall be oriented as shown on the applicable drawing.

4.6.7 Test methods.

4.6.7.1 Chemical composition. Chemical analysis shall be conducted in accordance with the applicable methods specified in ASTM A751.

4.6.7.2 Yield strength. Yield strength shall be determined by the offset method in accordance with ASTM E8/E8M. The limiting offset shall be 0.10 percent (0.001 inch per inch of gauge length). The rate of stress application when determining the yield strength shall not exceed 100,000 pounds per square inch per minute.

4.6.7.3 Reduction of area. Reduction of area shall be determined in accordance with ASTM E8/E8M.

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4.6.7.4 Charpy impact resistance. Charpy V-notch impact tests shall be conducted in accordance with ASTM E23. The temperature of the test specimen at impact shall be minus 40 °F ± 2 °F. The average impact value obtained from the tests of the three specimens under ASTM E23 and ASTM A370 for both the transverse (TL) and longitudinal (LT) orientations shall be equal to or greater than the minimum acceptance value shown in table II.

4.6.7.5 Hardness. Brinell or Rockwell hardness tests shall be conducted in accordance with ASTM E10 or ASTM E18, as applicable. All surface scale and decarburization shall be removed from locations where tests are made.

4.6.7.6 Weld repair. Each forging which has been repaired by welding shall be examined by standard methods such as magnetic particle inspection, radiographic inspection, dimensional checks, etc., to determine compliance with 3.6.2.

4.6.7.6.1 Weld repair radiography. When radiography is required on the drawing or contract for weld repairs, the weld repair shall be inspected to assure that the defect has been removed and meets Level 3 of ASTM E390 for both lack of fusion and penetration. No cracks are permissible.

4.6.7.7 Macroetch. Macroetching shall be conducted in accordance with ASTM E340.

4.6.7.8 Cleanliness. Cleanliness shall be conducted in accordance with AMS 2301.

4.6.7.9 Dimensions and surface roughness. Forging dimensions shall be measured or gaged by using standard methods using appropriate instruments. Surface roughness shall be examined visually; in case of dispute, a Government approved measuring instrument shall be used.

4.7 Interpretation of test results.

4.7.1 Rejection. Any forging which fails any one of the quality conformance inspections or any other required inspection specified in the contract or order shall be rejected.

4.7.1.1 Obvious defects. If rejection of a forging can be traced to the presence of obvious defects (e.g. non-metallic) in the failed test specimen, the test values obtained on that specimen may be disregarded subject to the approval of the contracting officer. Two wake bars (see 6.7.11) shall be substituted for the defective specimen. When the wake bars have been substituted, failure of the forging to meet specification requirements based on test results from the wake specimens shall be cause for rejection of the forging, whether or not the wake bars also exhibit defects.

4.7.1.2 Defective machined specimens, faulty test equipment, and faulty testing methods. If either a defectively machined test specimen, faulty test equipment, or faulty operation of the testing equipment results in the rejection of a forging, the test values obtained from such specimens may be disregarded subject to the approval of the contracting officer, and another specimen from the same forging location may be substituted for each such specimen.

4.7.1.3 Retest. When a test specimen fails to comply with the applicable requirements of this specification for reasons other than that specified in 4.7.1, retesting and resubmission of each rejected lot shall be allowed as follows: when no sampling plan is provided by the procuring activity (see 6.2) and where there is no evidence that indicates that the specimen was not

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representative of the lot material, and when the detail specification does not otherwise specify, at least two specimens shall be selected to replace each specimen that failed. All specimens selected for retest shall meet the requirements of the specification or the lot shall be subject to rejection.

4.7.1.4 Check tests. From any lot of forgings found acceptable under prescribed inspection, the Government may designate any forging as a check forging. Such a forging shall be tested at a facility to be designated by the Government, and shall be subjected to any or all tests specified herein. If the forging fails to comply with the requirements if the applicable engineering drawing and Table II, two additional forgings shall be tested. If both forgings comply with all requirements, the lot shall be accepted. If either forging fails to comply with all the requirements, the lot shall be rejected.

4.7.1.5 Reporting of tests. The results of all tests and retests, chemical and mechanical, made on each forging (see 6.7.1), its test coupon (see 6.7.7), and its test metal (see 6.7.8), shall be recorded and maintained. Results shall be provided to the Government procuring activity, the Contracting officer, or the Government representative as necessary (see also 3.13).

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.1.1 Packaging details. For additional considerations for packaging and packing steel forgings see 6.6.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

a. Grades A, A1, and B cover steel forgings for critical applications and highly stressed applications, formerly covered under MIL-S-45203 that is now cancelled, are superseded by this specification.

b. Grades A, A1, and B should never be used for forgings for gun tubes, breech rings, bomb bodies, recoil and recuperator pistons, cylinders, and some types of shells.

c. Grades C and D cover steel forgings for general purpose applications such, for example as gear blanks and yokes, formerly covered by MIL-S-13048, that is now cancelled and superseded by this specification.

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d. Gun tube forgings should be procured under MIL-S-46119.

e. Breech ring forgings should be procured under MIL-R-10185 (NOT FOR NEW DESIGN).

f. Forgings for armor or ballistic applications should be procured under MIL-DTL-13823.

g. This specification is not intended to be used in the procurement of bars and billets for reforging.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of the specification.

b. Grade of forging required (see 1.2.1).

c. Applicable drawing number(s), if required.

d. A first article and instructions for submission, if required (see 3.2).

e. Chemical composition, if specified (see 3.3.1).

f. Requirement for certificate of chemical analysis (see 3.3.3).

g Cleanliness of material, if required (see 3.7.3).

h. Yield strength, if required (see 3.5.5.1).

i. Reduction of area, if required (see 3.5.5.2).

j. Charpy V-notch impact resistance, if required (see 3.5.5.3).

k. Hardness, if required (see 3.5.5.4).

l. Macroetching requirements, if required (see 3.7.2).

m. Metallographic requirements, if required (see 3.8).

n. Magnetic particle inspection, if not required (see 3.9).

o. Heat treat condition, if other than annealed or normalized (see 3.6.1).

p. If repair welding of forgings is authorized (see 3.6.2, 4.6.5.1, and 4.6.7.6)

q. If radiographic inspection of repair welds is required (see 4.6.7.6.1)

r. When retest procedures other than as specified in 4.7.1.3 are required.

6.3 Inspection and quality assurance. Certification of a contractor's quality system or program is not required by the Department of Defense if ISO or ANSI/ASQC standards are used.

6.4 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below should be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or the purchase order requirements. Deliverable data required by this specification is cited in table IV:

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TABLE IV. Required deliverable data.

PARAGRAPH NO.	DATA REQUIREMENT TITLE	APPLICABLE DID No. 1/
3.3.3	Certification Data Report	DI-MISC-82386
3.6.2	Repair Welding Approval for Forgings	DI-FORG-81195A
3.6.2.1, 3.13	Chemical and Physical Properties for Forgings Analysis Report	DI-FORG-81196A

1/ Copies of data item descriptions related to this specification will be approved and listed as such in DOD 5010.12, Acquisition Management Systems and Data Requirements Control List (AMSDL). Copies of data item descriptions required by the contractors in connection with specific acquisition functions are available online at https://quicksearch.dla.mil/.

6.5 Identification of test metal not sampled. Test metal from forgings not sampled should be removed from the forging and identified with the forging number.

6.6 Packaging and Packing notes.

6.6.1 Preservation and packaging. Cleaning, drying, preservation and packaging should be in accordance with the manufacturer's commercial practice.

6.6.2 Packing. All forgings should be packed for shipment in a manner to ensure carrier acceptance and safe delivery at destination.

6.6.3 Large forgings. Large forgings having projections that could be damaged in handling should be crated.

6.6.4 Small forgings. Small rough forgings should be packed rigidly in boxes, or should be shipped in burlap bags.

6.6.5 Marking. Marking should be in accordance with MIL-STD-129 in addition to any special marking required in the contract or purchase order.

6.7 Definitions.

6.7.1 Forging. Forging for the purpose of this specification consists of a mass of metal of sufficient dimensional size and volume to provide both the required body of the forging from which a finished component is to be manufactured and the required integral test metal or prolongation.

6.7.2 Chemical analysis. Suggested ASTM instrumental methods that can be used for chemical analysis are E415, E282, E484 and E322. ASTM A751 should be consulted for a list of available methods.

6.7.3 Heat treat lot. A heat treat lot consists of two or more production forgings made from the same heat of steel and either simultaneously heat treated in a batch type furnace or consecutively heat treated in a continuous belt fed heat treatment furnace.

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6.7.4 Prolongation. A prolongation, for purposes of inspection and testing, is the same as test metal (see 6.7.8) and is destined as an integral extension beyond the body of the forging from which test specimens are to be taken for mechanical property tests. The diameter and mass distribution of the prolongation should not be less than those of the forging at a point coincident with the end of the forging from which it extends.

6.7.5 Retest. A retest is any test made after the results of the original test are known.

6.7.6 Supplies. Supplies includes one or more forgings.

6.7.7 Test coupon. A mass or masses of metal sectioned from the body of a forging including at least a portion of the critical areas for which required mechanical properties are to be met. The test coupon is of sufficient dimensional size to provide the necessary chemical and mechanical property test specimens.

6.7.8 Test metal. Test metal is additional metal which is integrally forged and integrally heat treated material external to the body of the forging. The test metal should be of sufficient size and volume to provide all necessary test specimens for chemical and mechanical property tests as required. The diameter and mass distribution of the test metal should not be less than those of the forging at a point coincident with the end of the forging from which it extends.

6.7.9 Value difference. Value difference is that difference between the average value of the test for each mechanical property attribute except hardness of the material test coupon and the average value of the test for the same mechanical property attribute except hardness of the respective test metal. The value difference may be plus or minus.

Example:

Value difference = (First article test coupon average value) - (First article test metal average value).

6.7.10 Adjusted value. An adjusted value is a value calculated by algebraically adding the value difference to the particular mechanical property attribute value of the test metal.

Example:

Adjusted value = (Production article test metal value) + (value difference)

6.7.11 Wake bar. A wake bar is a test specimen which is substituted for another test specimen which has been discarded because of the presence of local defects caused by or overlooked in the preparation of the specimen. The wake bar should be taken as close as possible to the site of the original test specimen.

6.8 Finish-machine contour. The contour of the finish-machined component including important fillets should be indicated on the forging drawing or on a separate drawing furnished by the contracting officer.

6.9 Metric units. When metric dimensions are required, units for inch, foot, foot-pounds, and feet per second may be converted to the metric equivalent by multiplying them by the following conversion factors listed in table V.

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TABLE V. Conversion factors.

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/sec	0.3048	=	meter per second (m/s)
Fahrenheit (°F)	(t°F - 32)/1.8 = t°C		Celsius (°C)

Note: Conversion factors can be associated with ASTM E380 entitled “Metric Practice Guide.”

6.10 Subject term (key word) listing.

Clevis	Hardware	Shackles
Components	Hitches	Shear pins

6.11 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - MR

Preparing activity:
Army - MR

Project FORG-2025-003

Review activities:
Army - AR, AT, TE
DLA - CQ

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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